

# MD SOHAIL AHMED

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## CAREER OBJECTIVE

Enthusiastic Electronics and Communication Engineering student with strong academic performance and hands-on experience in IoT-integrated systems and full-stack web development. Seeking opportunities to apply engineering and software skills to build efficient, real-world solutions.

## EDUCATION

B.Tech in Electronics and Communication Engineering

Anurag University | CGPA: 8.80

Diploma in Electronics and Communication Engineering

Government Polytechnic College, Masab Tank, Hyderabad | CGPA: 9.91

SSC (10th)

TMRS Boys-1, Mahabubnagar | CGPA: 9.8

## TECHNICAL SKILLS

- Programming: Python (Intermediate), C (Basic), Java (Basic)
- Core Subjects: Analog Communication, Electronic Circuits, Signals & Systems
- Technologies: IoT Systems, Full Stack Web Development (Basic)
- Tools & Platforms: MATLAB, Wokwi Simulator, MongoDB Atlas, Render
- Backend: Node.js, Express.js (Basic)
- Hardware: ESP32, HC-SR04 Ultrasonic Sensor
- Concepts: Object-Oriented Programming, REST APIs

## PROJECTS

### Smart Municipality Reporter (IoT + Full Stack Web Application) [LINK](#)

- Developed an end-to-end smart waste management system integrating IoT hardware with a cloud-based web application
- Implemented bin-level detection using ESP32 and ultrasonic sensor (simulated via Wokwi) to classify waste levels and trigger actions
- Designed and built REST APIs (6-8 endpoints) using Node.js and Express.js with MongoDB Atlas for real-time data storage
- Enabled automated complaint generation when waste exceeds threshold, reducing manual monitoring effort
- Built multi-role dashboards (Admin, Worker, Citizen) with authentication and workflow tracking using unique complaint IDs
- Successfully tested complete system flow (IoT → Backend → Web Interface) ensuring reliable real-time monitoring
- Led web application development and system integration in a team-based hackathon environment

### PWM-Based Smart Lighting System Visualizer (MATLAB Simulation) [LINK](#)

- Designed and simulated an adaptive lighting system using Pulse Width Modulation (PWM) in MATLAB
- Implemented dynamic brightness control based on ambient light, occupancy, and emergency scenarios
- Analyzed multiple real-world conditions (low/medium/high light, gas leakage, rockfall emergencies)
- Demonstrated energy-efficient operation compared to traditional constant lighting systems
- Contributed to system modeling, simulation, and performance analysis in a team environment

## ACHIEVEMENTS

- Secured 1st Prize in Quiz Competition at IEEE CAS Industry Forum (Anurag University)
- Team Lead in multiple national-level hackathons demonstrating leadership and coordination skills

## **CERTIFICATIONS & PARTICIPATIONS**

- Python Beginner Certification
  - MATLAB Onramp Certification
  - Hackatronics 2.0 (36-hour National Hackathon)
  - OpenAI x NxtWave Regional Buildathon
  - Agentathon 2025 – Largest Agentic AI Hackathon (GDG Hyderabad)
  - Builderthon Participation
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## **EXTRA-CURRICULAR ACTIVITIES**

National Cadet Corps (NCC)

- - Achieved 'A' Grade in C Certificate (2025)
  - - Achieved 'A' Grade in B Certificate (2024)
  - - Completed 10-day military training camp gaining discipline and field exposure
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## **LANGUAGES**

- English
  - Urdu
  - Telugu
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## **STRENGTHS**

- Strong problem-solving and analytical thinking
- Ability to integrate hardware and software systems
- Leadership experience in team-based projects and hackathons
- Quick learner with adaptability to new technologies